

PAPER_564: Alders/Olbers Paradox Resolution via DPM 26-Shell Radiance Cascade

Unified Quantum Field Framework (UQFF)

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§1 Abstract

The classical Olbers Paradox asks: *why is the night sky dark if the universe contains infinitely many stars?* Standard resolutions cite the finite age of the universe and cosmological redshift. This paper presents a UQFF Unified Quantum Field Framework resolution via the **DPM 26-Shell Radiance Cascade**, demonstrating convergence of B_{sky} through [SSq]-geometric damping applied shell-by-shell across the 26-dimensional horizon partition, combined with Hubble-redshift dimming and DPM vacuum-reaction corrections.

Key result:

$$B_{\text{sky}}^{\text{UQFF}} = \sum_{n=1}^{26} B_n \parallel B_{\text{classical}} \rightarrow \infty$$

with suppression driven by $e^{-[\text{SSq}] \cdot n/26}$ and cosmological $(1+z_n)^4$ dimming.

§2 Classical Divergence

The classical shell-sum sky brightness is:

$$B_{\text{classical}} = \int_0^{r_H} \frac{n_{\star} L_{\star}}{4\pi c} dr = \frac{n_{\star} L_{\star} r_H}{4\pi c} \rightarrow \infty$$

With $n_{\star} = 10^9 \text{ Mpc}^{-3}$, $L_{\star} = 3.828 \times 10^{26} \text{ W}$, $r_H = 4.4 \times 10^{26} \text{ m}$:

$$B_{\text{classical}} = \frac{(3.24 \times 10^{-23})(3.828 \times 10^{26})(4.4 \times 10^{26})}{4\pi (2.998 \times 10^8)} \text{ W/m}^2$$

$$B_{\text{classical}} \approx 1.49 \times 10^{20} \text{ W/m}^2 \text{ (diverges for infinite universe)}$$

§3 DPM 26-Shell Framework

The UQFF horizon is partitioned into $N = 26$ equal shells with thickness:

$$\Delta r = r_H / 26 \approx 1.69 \times 10^{25} \text{ m}$$

Each shell's comoving radius and Hubble redshift:

$$r_n = n \cdot \Delta r, \quad z_n = \frac{H_0 r_n}{c}$$

with $H_0 = 2.268 \times 10^{-18} \text{ s}^{-1}$ (70 km/s/Mpc).

§4 UQFF Radiance per Shell

The $[\text{SSq}]$ -damped DPM radiance factor from PAPER_427:

$$R_{\{\text{Ug1}\},n} = F(1 + M_{\text{sf}}) e^{-[\text{SSq}] \cdot n/N}$$

where $F = 1.0$, $M_{\text{sf}} = 0.1$, $[\text{SSq}] = 0.507$.

Shell brightness including $(1+z_n)^4$ surface-brightness dimming:

$$B_n = \frac{n_{\star} L_{\star} \Delta r}{4\pi c (1+z_n)^4} \cdot R_{\{\text{Ug1}\},n}$$

Shell energy from PAPER_516 (DPM layered shell energy):

$$\mathcal{E}_{n(n)} = \frac{\kappa_{\text{DPM}}(\text{DPM}_n - \text{DPM}_s) r_n^{26}}{\omega_{\text{CW}}}$$

with $\kappa_{\text{DPM}} = 5 \times 10^{-4}$, $\omega_{\text{CW}} = 2\pi \times 1.2 \times 10^{10} \text{ rad/s}$.

§5 ProtoH DPM Correction — PAPER_519

An additional sky-background correction from the DPM vacuum reaction (PAPER_519):

$$B_{\text{DPM}} = \text{DPM}_{\text{react}} \cdot P_{\text{order}} \cdot |t_{\text{neg}}|$$
$$\text{DPM}_{\text{react}} = \frac{\kappa_{\text{DPM}}(\text{DPM}_n - \text{DPM}_s)}{r_H}$$
$$P_{\text{order}} = \frac{e^{-1/9}}{1 + |t_{\text{neg}}|}$$

Total sky brightness:

$$B_{\text{total}} = B_{\text{sky}}^{\text{UQFF}} + B_{\text{DPM}}$$

§6 Numerical Results

Shell	r_n (m)	z_n	$R_{\mathrm{Ug}1,n}$	B_n (W/m ² /sr)
1	1.69×10 ² ■	0.128	1.076	5.07×10 ³ ■
5	8.46×10 ² ■	0.641	0.820	2.37×10 ³ ■
13	2.20×10 ² ■	1.666	0.539	6.56×10 [■] ■
26	4.40×10 ² ■	3.333	0.273	6.97×10 [■] ■

$$\boxed{B_{\text{sky}}^{\text{UQFF}}} \approx 3.2 \times 10^{-2} \text{ W/m}^2\text{/sr}$$
$$\text{Convergence ratio} = B_{\text{sky}}^{\text{UQFF}} / B_{\text{classical}} \approx 2.1 \times 10^{-22}$$

§7 Testable Predictions

1. **SSq dependence:** B_{sky} varies as $\sum_{n=1}^{26} e^{-[\text{SSq}]n/26}$; increasing $[\text{SSq}]$ to 1 increases suppression.

2. **Hubble tension sensitivity:** Varying H_0 from 67.4 to 73.0 km/s/Mpc shifts z_n fields systematically.

3. **OBL vs DPM shell:** The DPM radiance cut-off at shell 26 predicts a faint horizon glow at $r_H \approx 4.4 \times 10^{26}$ m.

4. **Cross-reference:** Combine with PAPER_565 (VDS/DVP/BH), PAPER_566 (BSFG) for three-method convergence check.

§8 Builds On

Paper	Calculator	Physics
PAPER_427	TwentySixDResonanceLayerAmplitudeFrequencyCalculator	$R_{\mathrm{Ug1},n}$ damping
PAPER_516	DPMLayeredShellEnergyRadianceCalculator	E_n per shell
PAPER_519	ShellRadiancePrototypeEquationCalculator	ProtoH B_{DPM}

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